

AI Integration Opportunities in the AfriCAN DANCE Study

Ancestry-Aware Colorectal Cancer Genomics & Drug Discovery
Technical Overview for Research Team & Collaborators — July 2026

1. Research Study Overview

What the Study Wants to Achieve

Colorectal cancer (CRC) does not affect all communities equally. African Americans have the highest rates of CRC incidence and death of any racial group in the United States. Yet nearly all of our understanding of which genes drive colorectal cancer, which drugs work against it, and which treatment guidelines doctors follow is based on data from people of European ancestry. That means the tools doctors use to diagnose and treat CRC were essentially built for white patients and may not work as well for everyone else.

This study, called **AfriCAN DANCE** (African Colorectal Cancer Network and Disparities And Cancer Epidemiology), wants to close that gap. It brings together CRC tumor samples from four populations across three continents:

- **African Americans (US)** — 300 patients
- **Non-Hispanic Whites (US)** — 100 patients (reference comparison)
- **Ghanaians (West Africa)** — 600 patients
- **Ethiopians (East Africa)** — 400 patients

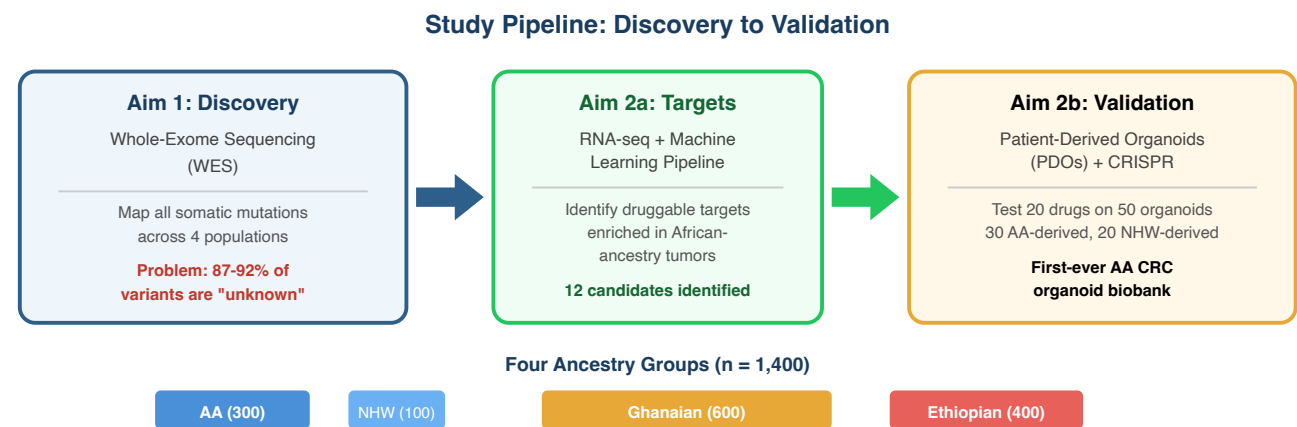


Figure 1. AfriCAN DANCE study pipeline. Tumor DNA and RNA flow from four populations through sequencing (Aim 1), computational drug target identification (Aim 2a), and wet-lab organoid validation (Aim 2b).

Stage 1 (Aim 1): Discover the Mutations

The researchers will read the complete protein-coding DNA from each patient's tumor and compare it to their healthy DNA to identify all the mutations present in the cancer. This process is called **Whole-Exome Sequencing (WES)**, which reads the roughly 2% of the genome that codes for proteins, where most cancer-driving mutations occur.

Early data already shows significant differences across populations. For example, the gene **APC** (a well-known CRC driver) is mutated in 93% of African American tumors but only 38% of Ghanaian tumors. The gene **PIK3CA** is mutated in 39.5% of African American tumors compared to just 3% of Ethiopian tumors. These differences suggest the molecular causes of CRC vary by ancestry.

The critical bottleneck: **87-92% of the mutations found are classified as "unknown significance."** When a mutation is found in a tumor, it gets categorized as harmful (pathogenic), harmless (benign), or unknown. A **Variant of Uncertain Significance (VUS)** means

there is not enough evidence to determine whether the mutation is driving the cancer or is just a harmless passenger. The classification tools currently in use (PolyPhen, SIFT) were trained primarily on European-ancestry data, creating a circular problem for African-ancestry variants.

Stage 2 (Aim 2a): Identify Drug Targets

DNA sequencing alone cannot determine whether a mutation affects tumor behavior. The researchers will also use **Ribonucleic Acid Sequencing (RNA-seq)**, which measures which genes are turned on or off and at what levels, to understand which mutations actually change the behavior of the tumor. A computational pipeline (currently using elastic net regression and random forest, two standard machine learning methods) will then nominate the most promising druggable targets. Early data shows that **MYC signaling** (a chain of molecular signals that drive tumor growth) is more active in African-ancestry CRC tumors. Twelve candidate drug targets have already been identified.

Stage 3 (Aim 2b): Validate in the Lab

Computational predictions need experimental proof. The team will build the **first-ever biobank of patient-derived organoids (PDOs) from African American CRC tumors**. PDOs are tiny 3D mini-tumors grown in the lab from actual patient cancer tissue. They closely mirror the genetics and behavior of the original tumor. Approximately 20 candidate drugs will be tested across 50 organoids. The team will also use **CRISPR** (a gene-editing technology that acts as molecular scissors for DNA) to introduce or remove specific mutations and confirm they cause differential drug response.

2. Where AI Can Assist: The 3 Recommendations

Recommendation 1: AI-Powered Variant Reclassification (Aim 1)

Effort: Low (2-3 weeks) | **Impact:** Very High

The Problem

The study's preliminary data shows that 87-92% of all genetic variants found in African-ancestry tumors cannot be classified as harmful or harmless. The tools currently used (PolyPhen and SIFT) are over a decade old and were trained primarily on European-ancestry data.

How AI Solves This

AlphaMissense (built by Google DeepMind) predicts whether a DNA change is harmful based on *protein structure and physical chemistry*, not clinical databases. It has pre-scored all 71 million possible single-letter DNA changes in the human genome. **EVE** (Evolutionary Model of Variant Effect) uses a different signal: how the same protein position has been conserved across species over millions of years. Neither tool requires the variant to appear in any patient database.

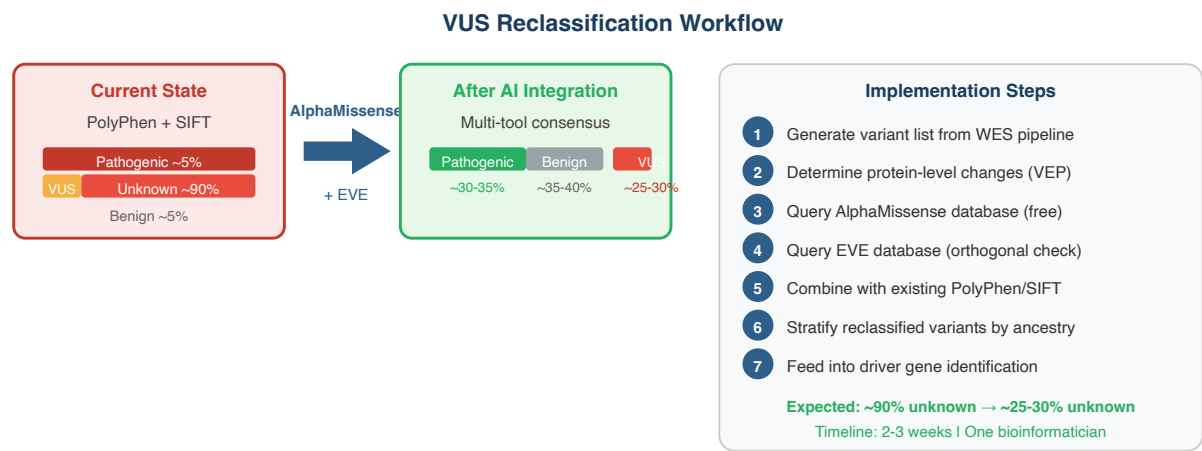


Figure 2. VUS reclassification workflow. Left: current state with ~90% unknown variants. Right: expected improvement after integrating AlphaMissense and EVE. Steps 1-7 detail the implementation pipeline.

Recommendation 2: AI-Improved DNA Variant Detection (Aim 1)

Effort: Low (1-2 weeks) | Impact: Moderate-High

The Problem

The standard human reference genome (GRCh38) was built primarily from European-ancestry samples. When sequencing DNA from African patients, this creates reference bias: some real variants get missed and some artifacts get incorrectly called as real. The variant caller currently planned (Mutect2, part of the Genome Analysis Toolkit or GATK) uses statistical formulas that struggle in repetitive and structurally variable genome regions.

How AI Solves This

DeepVariant (also from Google) replaces statistical rules with a **convolutional neural network (CNN)**, the same type of AI used for facial recognition. It converts sequencing data at each genome position into an image-like format, then uses a trained neural network to classify whether a variant is truly present. It has been independently shown to reduce false positives and false negatives by 15-30% in non-European samples.

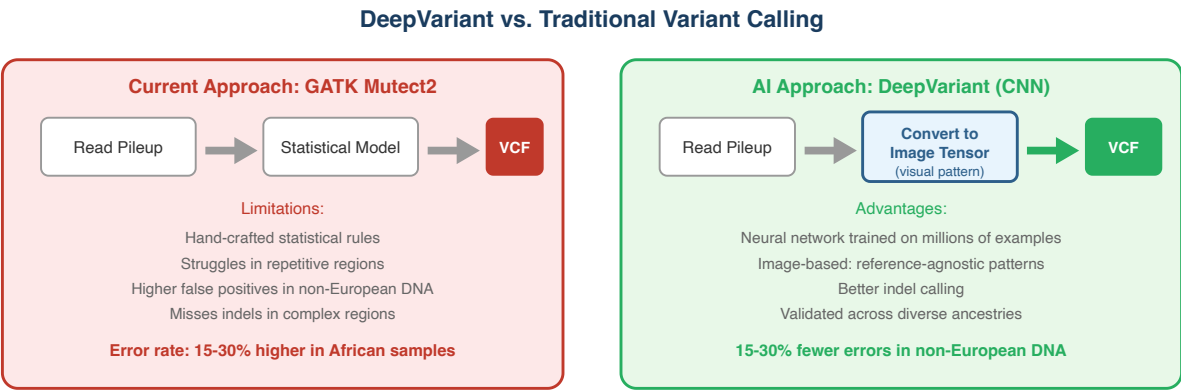


Figure 3. Comparison of traditional statistical variant calling (Mutect2) versus AI-based variant calling (DeepVariant). DeepVariant converts read pileups into image tensors and uses a convolutional neural network for classification.

Recommendation 3: AI Pre-Screening for Drug Response + AI Organoid Imaging (Aim 2b)

Effort: Moderate (2-3 months) | Impact: High

The Problem

The organoid drug screen tests ~20 drugs across 50 organoids at multiple concentrations over 14-day experiments. All 20 drugs enter the screen regardless of computational predictions. The current plan measures response with a single readout (CellTiter-Glo 3D viability assay), which only captures alive-versus-dead. Drug effects that alter organoid structure, invasion patterns, or differentiation state without killing the cell would be completely invisible.

How AI Solves This

Part A: AI models trained on large pharmacogenomic databases (GDSC, CCLE, DepMap) can predict drug response for each organoid before wet-lab work, prioritizing drugs most likely to show ancestry-specific effects.

Part B: Convolutional neural networks analyze microscope images of treated organoids to quantify structural changes beyond simple viability.

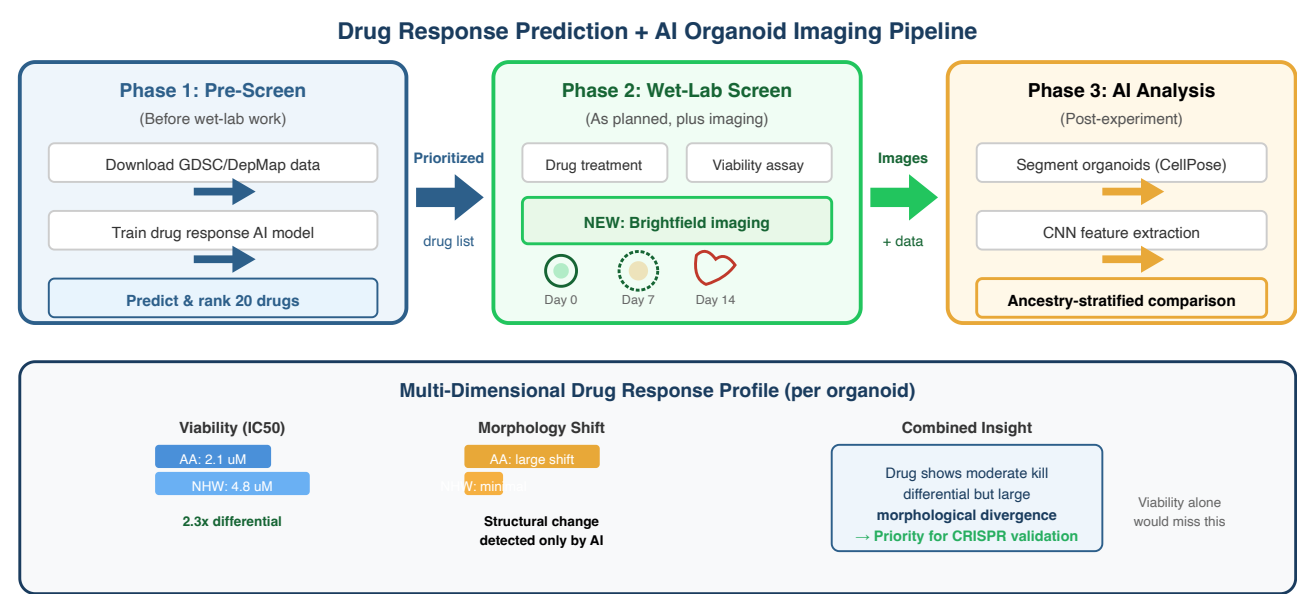


Figure 4. Three-phase drug screening pipeline. Phase 1: computational pre-screening prioritizes drugs. Phase 2: wet-lab organoid treatment with viability assay plus brightfield imaging. Phase 3: AI-based morphological analysis captures structural drug effects that viability assays miss.

3. Innovation Steps (Higher Effort, Higher Reward)

These go beyond applying existing AI tools. They involve building new AI systems that would represent genuinely novel scientific contributions suitable for separate methods publications.

Innovation 1: Custom Multi-Modal Variant Effect Predictor

Effort: 4-6 months | **Needs:** Dedicated ML engineer

Build a custom AI system that combines multiple data types to classify genetic variants specifically in the African-ancestry CRC context. AlphaMissense is general-purpose (any disease, any population). This study has something unique: matched somatic mutations, RNA-seq, clinical outcomes, and organoid drug response data from African-ancestry CRC tumors. That combination is globally unique.

The system would take in three types of information for each variant:

- **Genomic context:** variant position, type, conservation score across species
- **Structural context:** 3D protein location (from AlphaFold3), proximity to functional sites
- **Transcriptomic context:** whether tumors carrying the variant show abnormal gene expression (from RNA-seq data)

These feed into a neural network that outputs: pathogenicity probability, driver probability, and ancestry-enrichment flag. Training labels come from ClinVar/COSMIC known variants plus the study's own organoid validation data.

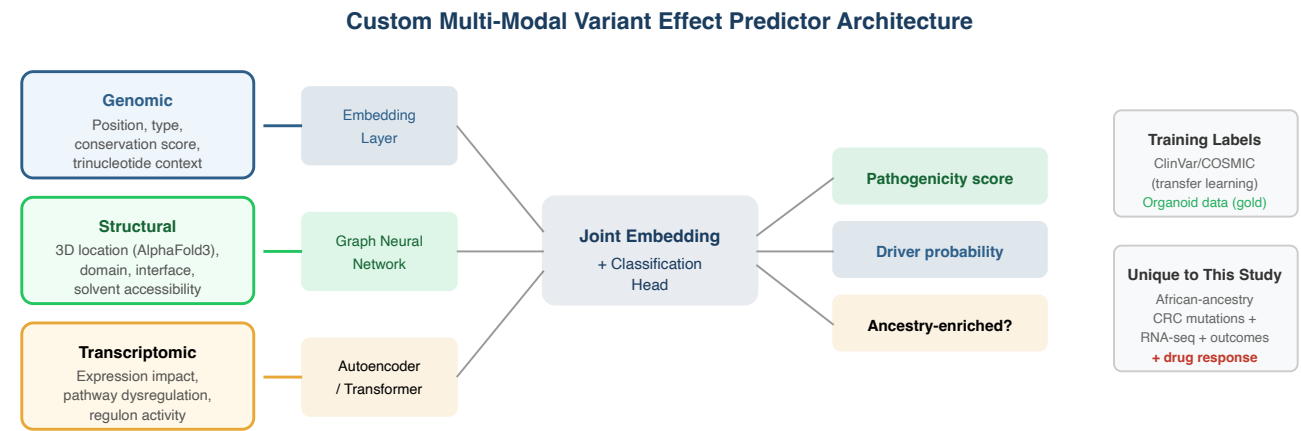


Figure 5. Proposed architecture for a custom multi-modal variant effect predictor. Three input branches (genomic, structural, transcriptomic) merge into a joint classifier. Training combines public databases with the study's unique African-ancestry CRC dataset.

Innovation 2: Network-Based AI for Drug Target Discovery

Effort: 3-4 months | **Needs:** Graph ML expertise

Replace single-gene analysis with a **Graph Neural Network (GNN)** that operates on protein-protein interaction networks. Genes do not work in isolation; they operate in interconnected networks like a social network. A single gene might look unremarkable when analyzed alone but could be a critical hub in a network neighborhood that is collectively rewired in African-ancestry tumors. Standard analysis misses these network-level patterns entirely.

The GNN would use a map of all known protein interactions (from databases like STRING or BioGRID) with each protein tagged with the study's mutation and expression data. The model identifies entire neighborhoods of the network that are collectively dysregulated in African-ancestry tumors, even if no single gene in that neighborhood stands out alone.

Innovation 3: Causal AI for Treatment Response

Effort: 3-5 months | **Needs:** Causal inference expertise

Apply causal inference machine learning to answer cause-and-effect questions rather than just prediction questions. Standard AI asks: "Can I predict drug response from tumor features?" Causal AI asks: "Is this specific mutation *causing* the different drug response, or is it just correlated with something else?"

Example: if KRAS G12D (more common in AA tumors) is associated with worse drug response, is that because KRAS G12D itself causes resistance, or because it co-occurs with other ancestry-linked features? Methods like Double Machine Learning can isolate the causal effect. The CRISPR experiments in Aim 2b provide exactly the experimental ground truth needed to validate these causal predictions.

4. Summary: Priority Matrix

Recommendation	Type	Effort	Timeline	Impact	Key Benefit
AlphaMissense/EVE (Aim 1)	Apply existing tools	Low	2-3 weeks	Very High	Reclassifies ~60-70% of "unknown" variants using protein structure AI, directly addressing the study's biggest bottleneck
DeepVariant (Aim 1)	Apply existing tools	Low	1-2 weeks	Moderate-High	Reduces false variant calls by 15-30% in African-ancestry DNA through image-based neural network detection
Drug pre-screen + organoid imaging (Aim 2b)	Build pipeline	Moderate	2-3 months	High	Predicts drug response before testing to prioritize experiments; captures morphological drug effects invisible to viability assays
Custom multi-modal variant predictor	Novel ML system	High	4-6 months	Very High	First ancestry-aware, CRC-specific AI variant classifier. Publishable as a novel method. Exploits the study's globally unique dataset
Network-based AI for drug targets	Novel ML system	High	3-4 months	High	Identifies drug targets by analyzing protein interaction networks rather than single genes, capturing network-level vulnerabilities
Causal AI for treatment response	Novel ML system	High	3-5 months	Very High	Determines whether mutations truly cause differential drug response vs. mere correlation. Tightens computational-to-experimental loop with CRISPR validation

Recommended Framing for Grant Revision

If integrating AI into the grant, the framing should be: **"AI-enhanced variant interpretation and drug response prediction to address the ancestry-specific knowledge gap."** The reviewer-friendly narrative is about closing the VUS gap and improving organoid screen efficiency, not about the algorithms themselves. The variant classification gap (87-92% unknown) is the most compelling, scientifically grounded argument for AI integration in this study.

The tiered approach (apply existing tools immediately, build custom systems for innovation) de-risks the project: if the custom model does not outperform existing tools, the Tier 1 results still stand.